Numerical Simulation of the Magnetic Field Around a Helmholtz Coil

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Introduction

Questions have been raised as to the potential biological effects of extremely low frequency electromagnetic fields (ELF-EMF) produced by electronic devices and power lines, however results vary as to what, if any, these effects are.

The goal was to model the magnetic field of these coils to place zebrafish embryos while testing the potential neurological effects of the field.

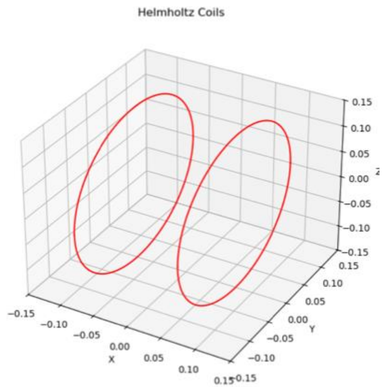


Figure 1. *a Helmholtz Coil configuration as two coils of wire separated by a distance equal to their radius.*

Methods

Using Python, coils were represented as small pieces of wire in the shape of a single loop, and the magnetic field was calculated at every point around the loops using the Biot-Savart Law (Eq. 1) where I is current in amperes, r is distance in meters.

$$B(r) = \frac{\mu_\theta}{4\pi}\int_C \frac{I\, dl \times r'}{|r'|^3} \qquad (1.)$$

Loops were divided into 50 pieces based on a convergence test. Based on Eq. 2, where R is radius, n is number of loops, a current of 138.59 amps was used to simulate a magnetic field of 0.829mT.

$$B = \left(\frac{4}{5}\right)^{\frac{3}{2}} \cdot \frac{\mu_\theta n I}{R} \qquad (2.)$$

Measurements of the field around the coils were taken out to 0.2 m using a magnetometer. Measurements were repeated five times on each axis and averaged.

Results

Accuracy was checked analytically and with measurements. The simulated value for the field at the center of the coils was found to be within 0.529% of the value calculated with equation 2. Along the x-axis, the simulated field was within 5% of the measured field out to 0.2m, on the y-axis the field was within 5% out to 0.1m.

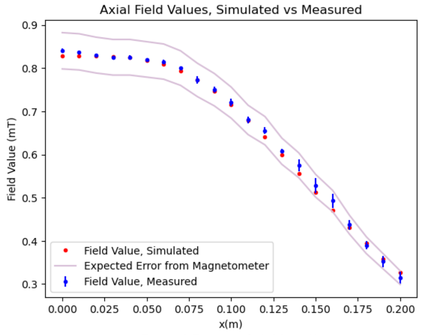


Figure 2.*. Error bars on measured values have been plotted but are generally too small to be visible.*

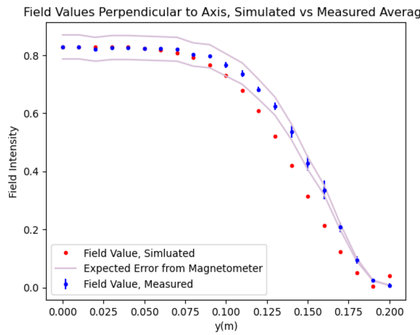


Figure 3. *Error bars on measured values have been plotted but are generally too small to be visible.*

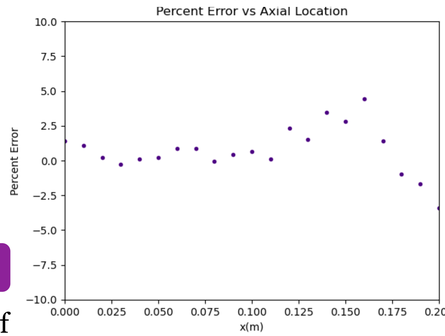


Figure 4. *Shows the error between simulated and measured results on the x-axis*

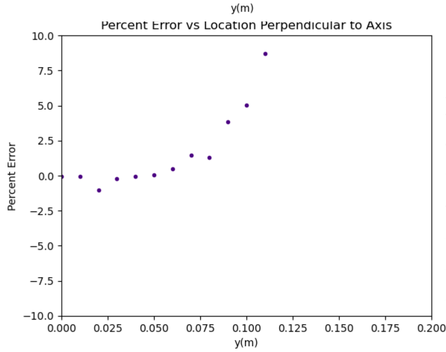


Figure 5 *Shows the error between simulated and measured results on the x-axis .*

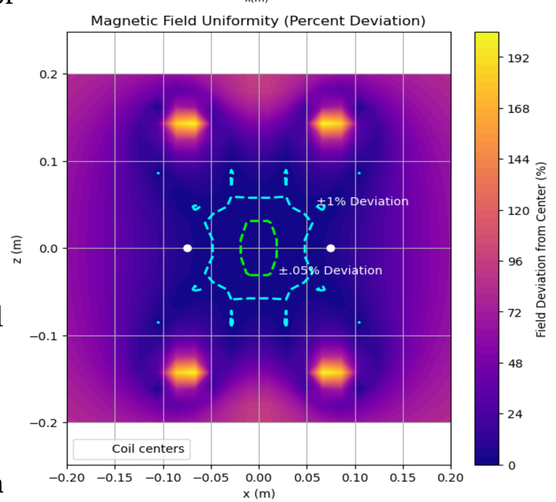


Figure 6. ***S****hows the percent deviation from center of the field around the coils. The yellow spots are the areas which the coils pass through.*

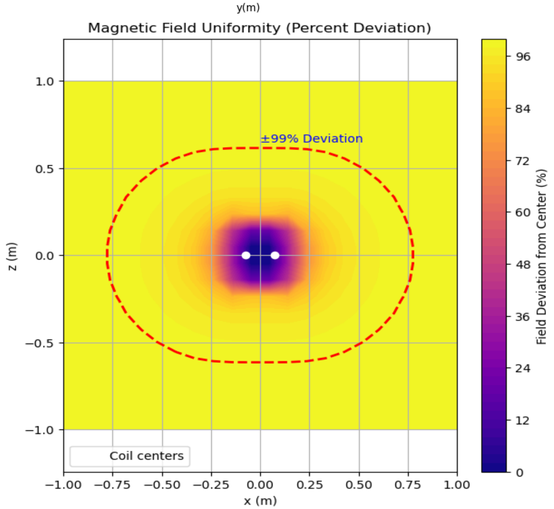


Figure 6. *shows a zoomed-out view of the heat map to the left with the region where the field is very small.*

Results Cont.

Some asymmetries are found in the data which are not predicted by the simulation.

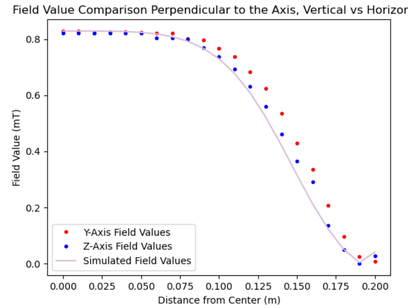


Figure 8. *Shows the differences between the y and z axes and the simulation.*

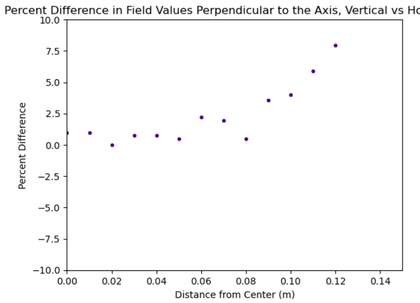


Figure 9. *Shows that the axes are within 5% different out to 0.1 m, within the area of interest for the experiment.*

Conclusion

With this simulation, we can better understand and predict the behavior of our field around the Helmholtz Coils, for more accurate experimentation.

In the future, we are interested in modeling an AC field for experiments with electromagnetic radiation.

References

1. *Chabay, Ruth, and Bruce A. Sherwood. Matter and Interactions. 4th ed., Conway, AR, Wiley, 2017.*
2. "12.5: Magnetic Field of a Current Loop." *Physics LibreTexts*, 2 March 2025, https://phys.libretexts.org. Accessed 30 June 2025.
3. ChatGPT was used for some plotting and code checking.

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